

Research Paper Outline

Name:	Ellen Rushe
Student Number:	12345678

Research Aims

Research Paper Title:	Determining the factors that dictate transport availability in Ireland.
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Research Questions/Tasks:	
1.	Is population density the strongest predictor of transport availability?
2.	What other features are associated with transport availability in a location?
3.	Which areas have the lowest transport options compared to their population density?

Dataset(s)

Copyright and data usage	
Dataset(s) source(s) (provide links to sources):	Dataset 1. PSRA –Dublin Presidential Property Price Register from 2021 Publisher: Property Services Regulatory Authority via: Open Data Unit, Department of Public Expenditure NDP Delivery and Reform, Government of Ireland URL: data.gov.ie (known henceforth as ODU)

	URL to download: https://data.gov.ie/dataset/dublin-residential-property-price-register
	Dataset 2: Public Transport Infrastructure Management System (PTIMS) Publisher: National Transport Authority via ODU. URL to download: https://data.gov.ie/dataset/public-transport-infrastructure-management-system-ptims
	Dataset 3: Railway Stations - National 250k Map Of Ireland Publisher: Tailte Éireann – Surveying via ODU. URL to download: https://data.gov.ie/dataset/railway-stations-national-250k-map-of-ireland2
	Dataset 4: Luas Stop Locations Publisher: Transport Infrastructure Ireland via ODU URL to download: https://data.gov.ie/en_GB/dataset/luas-stop-locations/resource/772ab84a-6ffd-48fb-8508-a175c74c2385
	Dataset 5: Population Distribution and Movements Publisher: Central Statistics Office via ODU. Dataset 6: https://data.gov.ie/dataset/population-classified-by-area

Licence(s):	Creative Commons Attribution 4.0 (CC-BY 4.0) licence (see [1]) Licence given by publisher on each data source link above and detailed in [2].
Terms of Licence (s):	1. Credit must be given. 2. A link to the CC-BY licence should be provided. 3. A statement on the changes made to the original material by the processor should be given. Additional information is given in [1].
Can data be used for non-commercial use cases?	Yes

GDPR	
Does data contain special Category data?	No the data does not contain any information about any individual falling into the categories outlined in Article 9 of the General Data Protection Regulation [4].
If yes, what are they?	N/A
If yes, on what grounds are you processing this data?	N/A

Ethical considerations

Outline of ethical considerations (max. 200 words)
It is intended to link data on transport, property prices and population distribution together in order to determine the factors that most impact transport availability. It is intended to determine whether factors other than population density are predictive of increased transport infrastructure within a particular area, which could shed light on inequalities in the distribution of public services across the country. Though this analysis is intended to illuminate inequalities within society, the linked dataset created by combining all data sources could facilitate the development of models to automatically assign prices to properties for sale. In particular, using location to determine house prices - rather than the structural properties (e.g. squared metreage) of a given property - serves to price lower-income members of society out of particular geographic areas due to the propagation of existing bias in house pricing data, which assigns higher prices to particular geographical locations [4].

Impact on analysis (max. 200 words)

The linked dataset, and the code to link it, will not be published publicly in order to minimise its availability for ethically dubious purposes.

This analysis should use interpretable machine learning models and analysis in order to understand the relationship between the transport infrastructure distribution and the factors analysed.

References

1. "Attribution 4.0 International CC BY 4.0 Deed". Creative Commons. Available at: <https://creativecommons.org/licenses/by/4.0/> [Accessed 25 June, 2024]
2. "The Open Data Licence". Open Data Unit, Government of Ireland. Available at: <https://data.gov.ie/pages/opendatalicence> [Accessed: 25 June, 2024]
3. Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation) [2016] OJ L 119, 4.5.2016, pp. 1–88. Available at: <https://eur-lex.europa.eu/eli/reg/2016/679/oj> [Accessed: 25 June, 2024]
4. Hochstenbach, Cody, and Rowan Arundel. "Spatial housing market polarisation: National and urban dynamics of diverging house values." *Transactions of the Institute of British Geographers* 45.2 (2020): 464-482.